

EHF-2025-May-PYQ Answers

Q1. [5 Marks each]

- (a) Write the difference between hacking and ethical hacking with examples.
- (b) Define digital forensics and state its primary purpose.
- (c) What is electronic evidence? Provide two examples.
- (d) Define forensic duplication and explain its importance in digital forensics.

Q2. [10 Marks each]

- (a) What are the various types of network attacks? How do these attacks impact a network?
- (b) Explain the process of mobile forensic investigation, from evidence collection to reporting.

Q3. [10 Marks each]

- (a) Describe in detail the structure and layout of an investigative report used in digital forensics.
- (b) Describe the role of computers in modern-day crimes with examples.

Q4. [10 Marks each]

- (a) Explain Chain of Custody in detail.
- (b) Write Challenges in mobile forensics.

Q5. [10 Marks each]

- (a) What are the challenges in network forensics? Explain any two tools used for network forensics.
- (b) Discuss the importance of writing an investigative report in forensic investigations. What goals does it aim to achieve?

Q6. [10 Marks each]

Write a short note on (Any Two):

1. Rules of ethical hacking and types of hackers.
2. Incident Response and Methodology of Incident Response.
3. Tools used in Computer forensics.

Q1. [5 Marks each] - Answers

a. Write the difference between hacking and ethical hacking with examples.

Aspect	Hacking	Ethical Hacking
Definition	Unauthorized access to computer systems or networks.	Authorized testing of systems to find and fix vulnerabilities.
Intent	Malicious: steal data, damage systems, disrupt services.	Protective: improve security, prevent attacks.
Legality	Illegal activity, punishable by law.	Legal activity, performed with permission.
Approach	Exploits weaknesses secretly.	Uses same techniques but reports findings responsibly.
Example	A hacker stealing credit card details from a website.	An ethical hacker performing penetration testing for a bank's online system.

b. Define digital forensics and state its primary purpose.

Definition of Digital Forensics

- Digital Forensics is the process of identifying, preserving, analyzing, and presenting digital evidence from electronic devices.
- It is used in investigations related to cybercrime, fraud, and unauthorized access.

Primary Purpose

- To collect and preserve digital evidence in a legally acceptable manner.

- To reconstruct events and understand how a crime or incident occurred.
- To support law enforcement and legal proceedings.
- To detect, prevent, and respond to cyberattacks.
- To ensure integrity and authenticity of evidence during investigation.

c. What is electronic evidence? Provide two examples.

Electronic Evidence – Definition

- Electronic evidence refers to any information or data stored or transmitted in digital form that can be used in a court of law.
- It includes files, emails, logs, or digital records obtained from computers, mobile devices, or networks.
- Must be collected, preserved, and presented following proper **chain of custody** to ensure admissibility.
- Can be either **primary evidence** (direct digital records) or **secondary evidence** (derived reports, screenshots).

Examples

1. **Email records** showing communication between parties in a fraud case.
2. **Browser history** showing visits to suspicious or illegal websites.

d. Define forensic duplication and explain its importance in digital forensics.

Definition of Forensic Duplication

- Forensic duplication is the process of creating an exact bit-by-bit copy of digital storage media (like hard drives, USBs, or memory cards).
- The duplicate is used for analysis while the original is preserved to maintain evidence integrity.

Importance in Digital Forensics

- Ensures the original evidence remains untouched and legally admissible.
- Allows investigators to perform repeated tests without risk of altering the source.
- Maintains chain of custody and authenticity of digital evidence.
- Helps in recovering deleted or hidden files during investigation.
- Provides a reliable copy for court presentation and expert analysis.

Q2. [10 Marks each] - Answers

a. What are the various types of network attacks? How do these attacks impact a network?

1. Definition

A **network attack** is any unauthorized activity aimed at disrupting, stealing, or damaging data and resources within a computer network. Attackers exploit vulnerabilities to gain access, cause downtime, or steal sensitive information.

2. Types of Network Attacks

1. Denial of Service (DoS) / Distributed DoS (DDoS)

- Flooding the network with traffic to make services unavailable.

2. Man-in-the-Middle (MITM)

- Attacker intercepts communication between two parties to steal or alter data.

3. IP Spoofing

- Using a fake IP address to impersonate a trusted system.

4. Phishing & Social Engineering

- Trick users into revealing credentials or sensitive data.

5. Password Attacks (Brute Force, Dictionary, Keylogging)

- Attempting to crack user credentials to gain unauthorized access.

6. Malware-based Attacks (Viruses, Worms, Trojans)

- Malicious software spreads through the network, damaging systems.

3. Impact of Network Attacks

- **Data Theft** – Loss of confidential information.
- **Service Disruption** – Downtime due to DoS/DDoS attacks.
- **Unauthorized Access** – Attackers gain control over systems.
- **Integrity Loss** – Altered or corrupted data.
- **Trust Issues** – Customers lose confidence in the organization.

4. Example

- A **DDoS attack** on a banking website floods servers with fake requests, making online banking unavailable to genuine users.

b. Explain the process of mobile forensic investigation, from evidence collection to reporting.

1. Definition

Mobile forensic investigation is the process of **recovering, preserving, analyzing, and presenting evidence** from mobile devices (smartphones, tablets, SIM cards, memory cards) in a legally acceptable manner.

2. Steps in the Process

1. Evidence Collection

- Seize the mobile device securely.
- Prevent remote access (e.g., airplane mode, Faraday bag).
- Document device details (IMEI, model, SIM info).

2. Preservation

- Ensure data integrity by creating forensic images (bit-by-bit copies).
- Maintain chain of custody records.

- Store original device safely to avoid tampering.

3. Extraction

- Use forensic tools (Cellebrite, Oxygen Forensics, XRY) to extract data.
- Collect call logs, SMS, emails, app data, multimedia, GPS, browser history.
- Handle encrypted or locked devices carefully.

4. Analysis

- Examine extracted data for evidence of crime (messages, files, location).
- Correlate findings with timelines and user activity.
- Detect deleted or hidden data using recovery tools.

5. Documentation

- Record every step: who handled the device, when, and how.
- Maintain logs of tools used and results obtained.

6. Reporting

- Prepare a clear forensic report with findings.
- Include screenshots, logs, and timelines.
- Present evidence in court with expert testimony.

Q3. [10 Marks each] - Answers

a. Describe in detail the structure and layout of an investigative report used in digital forensics.

Structure & Layout

1. Title Page

- Case name/number
- Investigator details
- Date of report

2. Executive Summary

- Brief overview of the case
- Key findings and conclusions

3. Introduction / Scope

- Purpose of investigation
- Scope and limitations
- Devices examined (e.g., mobile, computer, server)

4. Methodology

- Tools and techniques used (e.g., forensic imaging, analysis tools)
- Procedures followed to ensure integrity

5. Evidence Collection

- Details of seized devices
- Chain of custody documentation
- Imaging process and hash values

6. Analysis Section

- Examination of files, logs, emails, chats, multimedia, network data
- Timeline reconstruction of events
- Recovery of deleted/hidden data

7. Findings

- Clear presentation of evidence discovered
- Screenshots, tables, and logs supporting findings

8. Conclusion

- Summary of results
- Relevance of findings to the case

9. Recommendations (if applicable)

- Suggested security improvements or preventive measures

10. Appendices

- Supporting documents, tool logs, hash values, screenshots
- Glossary of technical terms

b. Describe the role of computers in modern-day crimes with examples.

1. Definition

Computers play a central role in **modern-day crimes** by acting as tools, targets, or mediums for illegal activities. Criminals exploit computer systems, networks, and digital platforms to commit offenses ranging from fraud to cyber terrorism.

2. Roles of Computers in Crimes

1. As a Tool

- Used to plan, execute, and coordinate crimes.
- Example: Hackers using malware to steal data.

2. As a Target

- Attacks aimed directly at computer systems and networks.
- Example: Denial of Service (DoS) attacks on servers.

3. As a Medium

- Facilitates communication and illegal transactions.
- Example: Dark web marketplaces for drugs or weapons.

3. Examples of Computer-Enabled Crimes

- **Cyber Fraud:** Online banking fraud, credit card scams.
- **Identity Theft:** Stealing personal data for impersonation.
- **Phishing Attacks:** Fake emails tricking users into revealing credentials.

Q4. [10 Marks each] - Answers

a. Explain Chain of Custody in detail.

1. Definition

The **Chain of Custody (CoC)** is the documented process that tracks the handling of digital evidence from the moment it is collected until it is presented in court. It ensures that evidence remains authentic, untampered, and legally admissible.

2. Key Elements

1. Collection

- Evidence is identified and seized (e.g., hard drives, mobile phones).
- Details like date, time, and collector's identity are recorded.

2. Preservation

- Evidence is stored securely to prevent alteration.
- Forensic imaging is used to create exact copies.

3. Transfer

- Every movement of evidence (from one investigator to another) is logged.
- Signatures, timestamps, and purpose of transfer are documented.

4. Analysis

- Forensic experts examine copies, not originals.
- Tools and methods used are recorded for transparency.

5. Presentation

- Evidence is presented in court with supporting documentation.
- Chain of Custody records prove authenticity and integrity.

b. Write Challenges in mobile forensics.

1. Definition

Mobile forensics deals with recovering and analyzing evidence from mobile devices. The process faces several challenges due to rapid technological changes and security features in smartphones.

2. Key Challenges

1. Device Diversity

- Wide range of mobile operating systems (Android, iOS, Windows) and hardware models.

2. Rapid OS & App Updates

- Frequent updates make forensic tools outdated quickly.

3. Data Encryption

- Strong encryption and security features hinder evidence extraction.

4. Volatile Data

- Temporary files, caches, and RAM data can be lost easily.

5. Cloud Storage Integration

- Data stored remotely in cloud services complicates access.

6. Legal & Privacy Issues

- Accessing personal data requires strict adherence to laws and warrants.

7. Anti-Forensic Techniques

- Attackers may use apps or methods to hide, delete, or obfuscate evidence.

Q5. [10 Marks each] - Answers

a. What are the challenges in network forensics? Explain any two tools used for network forensics.

1. Definition

Network forensics is the process of **capturing, storing, and analyzing network traffic** to investigate cyber incidents. It faces several challenges due to the dynamic and complex nature of modern networks.

2. Key Challenges

1. High Volume of Data

- Networks generate massive traffic; storing and analyzing all packets is difficult.

2. Real-Time Monitoring

- Attacks occur quickly; forensic tools must capture evidence instantly.

3. Encryption

- Widespread use of SSL/TLS and VPNs hides malicious activity.

4. Dynamic IPs & NAT

- Changing IP addresses and shared networks complicate attacker identification.

5. Distributed Attacks

- DDoS and botnet attacks originate from multiple sources, making tracing harder.

6. Data Volatility

- Network packets are transient and lost if not captured immediately.

7. Legal & Privacy Issues

- Monitoring traffic must comply with laws and user privacy rights.

3. Tools Used in Network Forensics

1. Wireshark

- Open-source packet analyzer.
- Captures and inspects live network traffic.
- Useful for detecting suspicious packets, malware communication, or protocol misuse.

2. tcpdump

- Command-line packet capture tool.
- Records network traffic for later analysis.

- Lightweight and widely used for quick forensic investigations.

b. Discuss the importance of writing an investigative report in forensic investigations. What goals does it aim to achieve?

1. Definition

An **investigative report** in digital forensics is a structured document that records the entire forensic process — from evidence collection to analysis and findings — ensuring clarity, transparency, and legal admissibility.

2. Importance of writing an investigative report in forensic investigations

1. Documentation of Process

- Records every step taken during investigation.
- Ensures transparency and accountability.

2. Evidence Integrity

- Maintains authenticity of digital evidence.
- Supports chain of custody requirements.

3. Legal Admissibility

- Courts accept evidence only if properly documented.
- Report acts as proof of proper handling.

4. Clarity for Stakeholders

- Provides investigators, lawyers, and judges with clear understanding.
- Summarizes technical findings in simple language.

5. Reference for Future Cases

- Serves as a knowledge base for similar investigations.
- Helps improve forensic methodologies.

3. Goals of an Investigative Report

- **Present Findings Clearly** – Evidence, analysis, and conclusions.
- **Support Legal Proceedings** – Provide admissible documentation in court.
- **Ensure Transparency** – Show that no tampering or bias occurred.
- **Facilitate Decision-Making** – Help law enforcement and judiciary act on evidence.
- **Provide Recommendations** – Suggest preventive or corrective measures.

Q6. [10 Marks each] - Answers

a. Write a short note on: Rules of Ethical Hacking and Types of Hackers

Definition

Ethical hacking is the authorized process of testing systems and networks to find security weaknesses before malicious hackers exploit them. Ethical hackers follow legal and professional rules while performing security testing.

Rules of Ethical Hacking

Ethical hackers must follow these rules:

1. Obtain Permission

- Hacking should be done only with proper authorization from the system owner.

2. Work Legally

- Follow cyber laws, company policies, and ethical standards.

3. Maintain Confidentiality

- Sensitive data found during testing must not be disclosed.

4. Do No Harm

- Avoid damaging systems, deleting data, or interrupting services.

5. Respect Privacy

- Personal and confidential information should not be misused.

Types of Hackers

1. White Hat Hackers (Ethical Hackers)

- Authorized professionals who test security legally.
- Example: Penetration testers.

2. Black Hat Hackers

- Unauthorized attackers who steal data or damage systems.

3. Gray Hat Hackers

- Hack without permission but may not have malicious intent.

4. Script Kiddies

- Beginners who use pre-made hacking tools without deep knowledge.

5. Hacktivists

- Hack for political or social causes.

6. State-Sponsored Hackers

- Work for governments for cyber espionage or attacks.

b. Write a short note on: Incident Response and Methodology of Incident Response

Definition

Incident Response (IR) is the process of identifying, handling, and recovering from security incidents such as malware attacks, data breaches, or unauthorized access.

Incident Response Methodology

Incident response generally follows six phases:

1. Preparation

- Develop policies and response plans.
- Train staff and form an incident response team.
- Install security tools like firewalls and IDS.

2. Identification

- Detect and verify that an incident has occurred.
- Analyze logs, alerts, and suspicious activities.

3. Containment

- Limit damage caused by the incident.
- Isolate infected systems from the network.

Types of containment:

- Short-term containment
- Long-term containment

4. Eradication

- Remove the root cause of the incident.
- Delete malware, close vulnerabilities, patch systems.

5. Recovery

- Restore affected systems and services.
- Monitor systems to ensure threats are removed.

6. Lessons Learned

- Review the incident.
- Prepare reports and improve future defenses.

Example

If ransomware infects a server:

- Identify infection
- Isolate the server
- Remove ransomware

- Restore files from backup
- Update security controls to prevent recurrence

c. Write a short note on: Tools Used in Computer Forensics

Definition

Computer forensic tools help investigators collect, preserve, analyze, and present digital evidence in a legally acceptable way.

Common Computer Forensic Tools

1. EnCase

- Popular digital forensic investigation tool.
- Used for disk imaging, evidence collection, and analysis.

2. FTK (Forensic Toolkit)

- Used for data recovery, email analysis, and password cracking.
- Fast searching and indexing of evidence.

3. Autopsy

- Open-source forensic tool.
- Used for analyzing disk images, deleted files, browser history.

4. Sleuth Kit

- Command-line forensic toolset.
- Helps examine file systems and recover evidence.

5. Wireshark

- Network forensic tool.
- Captures and analyzes network packets.

6. Cellebrite

- Used for mobile device forensics.
- Extracts deleted messages, call logs, and app data.

7. Helix

- Live forensic response tool.
- Used for incident response and evidence acquisition.

EHF-2025-December-PYQ Answers

Q1. [5 Marks each]

- a)** Define phishing attack and explain how attackers execute it.
- (b)** What is digital evidence? Discuss the admissibility criteria in court.
- (c)** Explain the role of Chain of Custody in digital investigation.
- (d)** Differentiate between Vulnerability Assessment and Penetration Testing.

Q2. [10 Marks each]

- (a)** Explain the Ethical Hacking lifecycle with a neat diagram.
- (b)** Describe the Incident Response Phases as per NIST framework.

Q3. [10 Marks each]

- (a)** Discuss Computer Forensics Investigation Process in detail.
- (b)** Explain how digital evidence is collected and preserved from Windows and Linux systems.

Q4. [10 Marks each]

- (a)** What are Network Forensics? Explain the tools used for analyzing network traffic.
- (b)** Describe E-mail Forensics. Explain steps to track header information and trace sender identity

Q5. [10 Marks each]

- (a)** Explain various Mobile Forensic acquisition techniques (Logical, Physical, Cloud-based).
- (b)** Elaborate common types of cybercrimes with suitable real-world examples.

Q6. [10 Marks each]

Write a short note on (Any Two):

1. **CSIRT** Roles and Responsibilities.
2. **Anti-Forensic** Techniques.
3. **Denial of Service (DoS)** Attack.
4. **Report Writing Guidelines** in Forensic Investigation

Q1. [5 Marks each] - Answers

a. Define phishing attack and explain how attackers execute it.

Definition of Phishing Attack

- Phishing is a cyberattack where attackers trick users into revealing sensitive information (like passwords, credit card numbers) by posing as a trusted entity.
- It usually happens through fake emails, websites, or messages.

Execution by Attackers

1. **Crafting Fake Communication** – Attackers send emails or messages that look genuine (e.g., from a bank or service provider).
2. **Deceptive Links** – The message contains links to fraudulent websites designed to mimic legitimate ones.
3. **Data Collection** – Victims enter personal details on the fake site, which attackers capture.
4. **Exploitation** – Stolen information is used for identity theft, financial fraud, or unauthorized access.
5. **Social Engineering** – Attackers exploit human trust and urgency (e.g., "Your account will be blocked, click here").

b. What is digital evidence? Discuss the admissibility criteria in court.

Definition of Digital Evidence

- Digital evidence is information stored or transmitted in digital form that can be used in investigations or legal proceedings.
- It includes data from computers, mobile devices, networks, or digital storage media.

Admissibility Criteria in Court

1. **Authenticity** – Evidence must be proven genuine and not altered.
2. **Integrity** – Chain of custody must be maintained to show it is untampered.
3. **Relevance** – Evidence must be directly related to the case.
4. **Reliability** – Methods of collection and analysis must be scientifically valid.
5. **Legality** – Evidence must be obtained through lawful procedures.

c. Explain the role of Chain of Custody in digital investigation.

Role of Chain of Custody in Digital Investigation

- Chain of Custody refers to the documented process that tracks the handling of digital evidence from collection to presentation in court.
- It ensures that evidence is properly preserved, transferred, and analyzed without tampering.

Key Roles

1. **Maintains Integrity** – Confirms evidence is unchanged and authentic.
2. **Provides Documentation** – Records who collected, handled, and analyzed the evidence.
3. **Ensures Accountability** – Each handler is responsible for preserving evidence correctly.
4. **Supports Legal Admissibility** – Courts accept evidence only if chain of custody is clear.

5. **Prevents Tampering/Contamination** – Protects against claims of manipulation or mishandling.

d. Differentiate between Vulnerability Assessment and Penetration Testing.

Aspect	Vulnerability Assessment	Penetration Testing
Definition	Process of identifying, classifying, and prioritizing weaknesses.	Simulated attack to exploit vulnerabilities in a system.
Objective	To detect and list potential security flaws.	To test how attackers can exploit those flaws.
Approach	Broad, automated scanning and analysis.	Manual + automated exploitation techniques.
Outcome	Report of vulnerabilities with severity ratings.	Proof-of-concept showing actual impact of exploitation.
Example	Using a scanner to find outdated software versions.	Attempting to break into a web app using SQL injection.

Q2. [10 Marks each] - Answers

a. Explain the Ethical Hacking lifecycle with a neat diagram.

1. Definition

The **Ethical Hacking lifecycle** is a systematic process followed by ethical hackers to test and strengthen security. It simulates real-world attacks in a legal, authorized manner to identify vulnerabilities before malicious hackers exploit them.

2. Phases of Lifecycle

1. Reconnaissance (Information Gathering)

- Collect information about the target (IP addresses, domains, DNS records).
- Types: Passive (public sources) and Active (direct probing).

2. Scanning

- Identify live hosts, open ports, and services.
- Tools: Nmap, Nessus.
- Purpose: Detect potential entry points.

3. Gaining Access

- Exploit vulnerabilities to enter the system.
- Tools: Metasploit, SQL injection scripts.
- Goal: Demonstrate how attackers could compromise systems.

4. Maintaining Access

- Install backdoors or create persistence to show long-term risks.
- Used to test how attackers can remain undetected.

5. Reporting

- Document findings, exploited vulnerabilities, tools used, and recommendations.
- Provides actionable steps for organizations to fix weaknesses.



b. Describe the Incident Response Phases as per NIST framework.

Incident Response Phases (NIST SP 800-61 Framework)

1. Preparation

- Establish policies, procedures, and incident response plans.
- Train the Computer Security Incident Response Team (CSIRT).
- Deploy monitoring tools (SIEM, IDS/IPS, endpoint detection).
- Ensure communication channels and escalation paths are defined.
- Maintain backups and test recovery procedures.

2. Detection & Analysis

- Identify potential incidents through alerts, logs, and anomaly detection.
- Differentiate between false positives and real threats.
- Classify incidents based on severity and impact.
- Document indicators of compromise (IoCs).
- Example: Detecting unusual outbound traffic indicating possible malware infection.

3. Containment, Eradication, and Recovery

- **Containment:** Limit the spread of the attack (e.g., isolate affected systems).
- **Eradication:** Remove malicious code, disable compromised accounts, patch vulnerabilities.
- **Recovery:** Restore systems to normal operation, validate integrity, monitor for recurrence.
- Example: Isolating infected servers during a ransomware attack, then restoring clean backups.

4. Post-Incident Activity

- Conduct a **lesson learned** meeting.

- Update incident response playbooks and security controls.
- Document what worked and what failed.
- Share findings with stakeholders and, if required, regulatory bodies.
- Example: After a phishing attack, organizations may improve email filtering and employee awareness training.

Q3. [10 Marks each] - Answers

a. Discuss Computer Forensics Investigation Process in detail.

1. Definition

Computer forensics is the practice of **collecting, preserving, analyzing, and presenting digital evidence** from computer systems in a way that is legally admissible. The investigation process ensures integrity and reliability of evidence.

2. Steps in the Process

1. Identification

- Recognize potential sources of digital evidence (hard drives, emails, logs).
- Define scope of investigation.

2. Preservation

- Secure and isolate the computer system.
- Create forensic images (bit-by-bit copies).
- Maintain chain of custody records.

3. Collection

- Extract data using forensic tools (EnCase, FTK).
- Gather files, logs, emails, browser history, registry entries.

4. Examination

- Filter and recover relevant data.
- Detect hidden, deleted, or encrypted files.

- Timeline reconstruction of user/system activity.

5. Analysis

- Interpret evidence to establish facts.
- Correlate findings with case objectives.
- Identify suspects, motives, and methods.

6. Documentation

- Record all steps, tools used, and findings.
- Ensure transparency and reproducibility.

7. Presentation

- Prepare a clear investigative report.
- Present evidence in court with expert testimony.

b. Explain how digital evidence is collected and preserved from Windows and Linux systems.

Windows Systems

1. Collection

- Acquire forensic image of hard drive using tools (FTK Imager, EnCase).
- Collect registry files (user activity, installed programs).
- Extract event logs (system, security, application logs).
- Gather browser history, temporary files, and user documents.

2. Preservation

- Generate hash values (MD5/SHA-1) for forensic images.
- Store evidence in write-protected media.
- Maintain chain of custody records.

Linux Systems

1. Collection

- Create bit-by-bit disk image using `dd` or `dcfldd`.
- Collect system logs (`/var/log/`), configuration files, and shell history.
- Extract user files, emails, and process information.
- Capture running processes and network connections using commands (`ps` , `netstat`).

2. Preservation

- Use cryptographic hash functions to verify integrity of images.
- Store original evidence securely; analyze only forensic copies.
- Document every step for accountability.

Q4. [10 Marks each] - Answers

a. What are Network Forensics? Explain the tools used for analyzing network traffic.

1. Definition

Network forensics is the branch of digital forensics that focuses on **capturing, storing, and analyzing network traffic** to investigate cyber incidents, detect intrusions, and gather legally admissible evidence.

2. Tools for Analyzing Network Traffic

1. Wireshark

- Open-source packet analyzer.
- Captures live network traffic and provides detailed protocol analysis.
- Useful for detecting suspicious packets, malware communication, or protocol misuse.

2. tcpdump

- Command-line packet capture tool.
- Records raw network traffic for later analysis.
- Lightweight and widely used for quick forensic investigations.

3. Snort

- Open-source Intrusion Detection System (IDS).
- Monitors network traffic in real time.
- Detects attacks using signature-based rules.

4. NetFlow Analyzer

- Collects and analyzes flow data from routers/switches.
- Helps identify unusual traffic patterns, bandwidth misuse, or DDoS attempts.

5. Xplico

- Network forensic analysis tool (NFAT).
- Reconstructs application-level data (emails, VoIP, HTTP sessions) from captured traffic.

b. Describe E-mail Forensics. Explain steps to track header information and trace sender identity

1. Definition

E-mail forensics is the process of examining e-mails to detect fraud, cybercrime, or misuse. It focuses on analyzing the content, metadata, and headers to trace the sender's identity and verify authenticity.

2. Steps to Track Header Information

1. Access Full Header

- Open the e-mail and view the complete header (contains routing details).

2. Analyze "Received" Fields

- Each mail server adds a "Received" line.
- The earliest entry (bottom-most) shows the original sender's IP address.

3. Check "From" and "Reply-To" Fields

- Compare with the actual sending server.
- Detect spoofing if mismatch occurs.

4. Inspect "Message-ID"

- Unique identifier generated by the sending mail server.
- Helps confirm authenticity.

5. Look at "Return-Path"

- Shows the actual return address used by the sender.

3. Steps to Trace Sender Identity

1. Identify Sender's IP Address

- Extracted from the earliest "Received" field.

2. Perform IP Lookup

- Use WHOIS or geolocation tools to find ISP and location.

3. Correlate with Logs

- Match IP with server logs, login records, or timestamps.

4. Verify Against Spoofing

- Check if the IP belongs to a legitimate mail server or attacker-controlled system.

Q5. [10 Marks each] - Answers

a. Explain various Mobile Forensic acquisition techniques (Logical, Physical, Cloud-based).

1. Definition

Mobile forensic acquisition refers to the process of extracting data from mobile devices in a forensically sound manner. Different techniques are used depending on the device type, operating system, and level of access required.

2. Techniques

1. Logical Acquisition

- Extracts data using the mobile OS's synchronization or backup features.
- Collects accessible files: contacts, SMS, call logs, media, app data.
- Tools: Cellebrite, Oxygen Forensics.
- **Limitation:** Cannot recover deleted or hidden files.

2. Physical Acquisition

- Creates a bit-by-bit copy of the entire device memory (including deleted data).
- Provides access to system files, hidden partitions, and unallocated space.
- Tools: XRY, UFED Physical Analyzer.
- **Advantage:** More comprehensive, includes deleted data.

3. Cloud-Based Acquisition

- Extracts data stored in cloud services linked to the mobile (Google Drive, iCloud).
- Collects synced contacts, messages, photos, app backups.
- Requires proper legal authorization (warrants).
- **Limitation:** Dependent on network access and credentials.

3. Example

- In a fraud case:
 - **Logical acquisition** retrieves SMS and call logs.
 - **Physical acquisition** recovers deleted WhatsApp chats.

- **Cloud acquisition** extracts synced Google Drive documents.

b. Elaborate common types of cybercrimes with suitable real-world examples.

1. Definition

Cybercrime refers to illegal activities carried out using computers, networks, or digital devices. These crimes exploit vulnerabilities in technology to steal, disrupt, or cause harm.

2. Types of Cybercrimes

1. Identity Theft

- Stealing personal information (SSN, bank details) to impersonate victims.
- *Example:* In 2017, the **Equifax data breach** exposed sensitive data of 147 million people.

2. Phishing Attacks

- Fake emails/websites trick users into revealing credentials.
- *Example:* The **Google Docs phishing scam (2017)** tricked millions into granting access to attackers.

3. Financial Fraud

- Online banking fraud, credit card scams, ransomware demanding payments.
- *Example:* The **WannaCry ransomware attack (2017)** affected organizations worldwide, demanding Bitcoin payments.

4. Child Exploitation & Abuse

- Distribution of illegal content via online platforms.
- *Example:* International law enforcement operations have shut down dark web child exploitation rings.

5. Corporate Espionage

- Stealing trade secrets or confidential data from companies.

- *Example:* The **SolarWinds cyberattack (2020)** compromised multiple government and corporate networks.

6. Cyberstalking & Harassment

- Using digital platforms to threaten or harass individuals.
- *Example:* Cases of online harassment via social media leading to legal action.

Q6. [10 Marks each] - Answers

a. Write a short note on: CSIRT Roles and Responsibilities.

1. Definition

A **Computer Security Incident Response Team (CSIRT)** is a specialized group that manages cybersecurity incidents. Its **roles** define what the team *is expected to do*, while its **responsibilities** explain *how they carry it out*.

2. Roles of CSIRT

1. **Monitoring Role** – Continuously watch systems and networks for threats.
2. **Analytical Role** – Investigate incidents, determine scope and impact.
3. **Defensive Role** – Contain and eradicate malicious activity.
4. **Restorative Role** – Recover systems and ensure business continuity.
5. **Advisory Role** – Provide guidance, awareness, and training to staff.
6. **Reporting Role** – Document incidents and communicate findings to management/legal bodies.

3. Responsibilities of CSIRT

- **Incident Detection** – Identify suspicious activity.
- **Incident Analysis** – Examine logs, malware, and attack vectors.
- **Containment & Eradication** – Isolate affected systems and remove threats.

- **Recovery** – Restore services securely.
- **Documentation & Reporting** – Maintain records for compliance and future prevention.
- **Awareness & Training** – Educate employees to reduce risks.

b. Write a short note on: Anti-Forensic Techniques.

1. Definition

Anti-forensic techniques are methods used by attackers to **hide, alter, or destroy digital evidence** in order to mislead investigators and obstruct forensic analysis. They aim to reduce the chances of detection and weaken the credibility of evidence.

2. Common Techniques

1. Data Hiding

- Concealing files using steganography, hidden partitions, or alternate data streams.

2. Data Obfuscation

- Encrypting or compressing files to make analysis difficult.

3. Data Deletion / Wiping

- Securely erasing files or overwriting disk sectors to prevent recovery.

4. Trail Obfuscation

- Modifying logs, timestamps, or metadata to mislead investigators.

5. Use of Anti-Forensic Tools

- Specialized software designed to erase traces or corrupt forensic tools.

c. Write a short note on: Denial of Service (DoS) Attack.

1. Definition

A **Denial of Service (DoS) attack** is a malicious attempt to make a computer system, network, or service unavailable to legitimate users by overwhelming it with excessive requests or exploiting vulnerabilities.

2. Key Features

- **Flooding Traffic** – Attackers send massive fake requests to exhaust resources.
- **Service Disruption** – Genuine users cannot access the service.
- **Resource Exhaustion** – CPU, memory, bandwidth, or server capacity gets overloaded.
- **Variants** – Distributed DoS (DDoS) uses multiple compromised systems (botnets).

3. Impact

- **Downtime** – Websites or services become inaccessible.
- **Financial Loss** – Organizations lose revenue and incur recovery costs.
- **Reputation Damage** – Customers lose trust in the service provider.
- **Security Risks** – Attack may be used as a distraction for other intrusions.

4. Example

- In 2016, the **Dyn DNS DDoS attack** disrupted major websites like Twitter, Netflix, and PayPal by flooding servers with traffic from IoT botnets.

d. Write a short note on: Report Writing Guidelines in Forensic Investigation

1. Definition

In forensic investigations, a **report** is a formal document that records the entire process of evidence collection, analysis, and findings. It ensures **clarity, transparency, and legal admissibility** of digital evidence.

2. Report Writing Guidelines

1. Clarity & Simplicity

- Use clear, concise language.

- Avoid technical jargon when addressing non-technical stakeholders.

2. Accuracy & Completeness

- Document every step of the investigation.
- Include tools used, methods applied, and evidence collected.

3. Objectivity

- Present facts without bias or personal opinions.
- Ensure neutrality in conclusions.

4. Structure & Organization

- Title page → Executive summary → Introduction → Methodology → Evidence → Analysis → Findings → Conclusion → Recommendations → Appendices.

5. Evidence Integrity

- Maintain chain of custody details.
- Include hash values to prove authenticity of digital evidence.

6. Legal Compliance

- Ensure the report meets admissibility standards in court.
- Follow organizational and jurisdictional guidelines.

7. Professional Presentation

- Use formal formatting, numbering, and references.
- Ensure readability for investigators, lawyers, and judges.

EHF-2024-May-PYQ Answers

Q1. [5 Marks each]

- (a) Explain Hackers, Crackers, Phreakers.
- (b) Explain the role of the Computer Security Incident Response Team (CSIRT).
- (c) What are different forensic image formats?
- (d) What is a Redundant Array of Inexpensive Disks (RAID)?

Q2. [10 Marks each]

- (a) Describe the method for recovering deleted files from a Windows system.
- (b) Write about malware handling in detail.

Q3. [10 Marks each]

- (a) What is an Intrusion Detection System (IDS)? Explain.
- (b) What is a file system and explain forensic analysis of a file system.

Q4. [10 Marks each]

- (b) Explain in detail the task of a hardware forensic tool.

Q5. [10 Marks each]

- (a) Write a short note on different types of scanning tools.
- (b) Explain the different types of digital forensics.

Q6. [10 Marks each]

- (b) Write a short note on (Any Two):

1. Illustrate investigation of a web browser.
2. Describe forensic duplication of a hard drive.
3. Explain the steps in router investigation.

Q1. [20 Marks] - Answers

a. Explain Hackers, Crackers, Phreakers.

Hackers

- Individuals skilled in computer systems and networks.
- Can be **ethical (white-hat)** or **malicious (black-hat)**.
- Example: A white-hat hacker performing penetration testing for a company.

Crackers

- Malicious users who break into systems to steal, damage, or misuse data.
- Focus on bypassing security protections like passwords or licenses.
- Example: Cracking software to remove license restrictions.

Phreakers

- Attackers who exploit telephone systems and networks.
- Originally targeted telecom lines to make free calls or manipulate billing.
- Example: Using special tones to bypass telephone billing systems.

b. Explain the role of the Computer Security Incident Response Team (CSIRT).

Role of CSIRT (Computer Security Incident Response Team)

- CSIRT is a specialized team responsible for handling and managing cybersecurity incidents in an organization.
- It ensures quick detection, response, and recovery from security breaches.

Key Roles

1. **Incident Detection** – Identifies security threats and suspicious activities.
2. **Incident Analysis** – Investigates the scope, impact, and cause of the incident.
3. **Containment & Eradication** – Prevents further damage and removes malicious elements.

4. **Recovery** – Restores systems and services to normal operation.
5. **Documentation & Reporting** – Maintains records for legal, compliance, and future prevention.
6. **Awareness & Training** – Educates employees to reduce risks and improve security posture.

c. What are different forensic image formats?

Forensic Image Formats

In digital forensics, evidence is preserved using specific image formats that store bit-by-bit copies of storage media. Common formats include:

1. Raw Format (.dd / .img)

- Simple bit-by-bit copy of the disk.
- Widely supported but large in size.

2. Expert Witness Format (EWF / .E01)

- Proprietary format used by forensic tools like EnCase.
- Supports compression, metadata, and integrity checks.

3. Advanced Forensic Format (AFF)

- Open-source format with flexibility.
- Allows compression, encryption, and metadata storage.

4. Generic Forensic Image (GFI)

- Used by some forensic suites.
- Provides structured storage with metadata.

5. Proprietary Tool Formats

- Formats specific to tools (e.g., FTK Imager).
- Include additional features like indexing or case notes.

d. What is a Redundant Array of Inexpensive Disks (RAID)?

Definition of RAID

- RAID (Redundant Array of Inexpensive Disks) is a storage technology that combines multiple physical hard drives into a single logical unit.
- It improves performance, provides fault tolerance, or both, depending on the RAID level used.

Key Points

1. **Redundancy** – Data is duplicated across drives to prevent loss.
2. **Performance** – Multiple drives can read/write simultaneously, increasing speed.
3. **Fault Tolerance** – Some RAID levels allow recovery if a drive fails.
4. **Levels** – Common levels include RAID 0 (striping), RAID 1 (mirroring), RAID 5 (striping with parity).
5. **Use Case** – Widely used in servers, data centers, and forensic investigations to ensure data reliability.

Q2. [10 Marks each] - Answers

a. Describe the method for recovering deleted files from a Windows system.

Method for Recovering Deleted Files from Windows

1. Preserve the System

- Immediately secure the system to prevent overwriting of deleted data.
- Use **write blockers** when accessing storage media.
- Maintain chain of custody documentation.

2. Create Forensic Image

- Generate a **bit-by-bit copy** of the hard drive using tools like **FTK Imager** or **EnCase**.

- This ensures the original evidence remains untouched.
- Verify integrity using hash values (MD5/SHA-1).

3. File System Analysis

- Windows uses **NTFS/FAT file systems**.
- Deleted files remain in unallocated space until overwritten.
- Forensic tools scan this space to locate remnants of deleted files.

4. Use Recovery Tools

- **EnCase, FTK, Autopsy, X-Ways Forensics** can reconstruct deleted files.
- These tools analyze Master File Table (MFT) entries and metadata.
- Carving techniques are used to recover files based on headers and footers.

5. Recover Metadata

- Extract file attributes (timestamps, size, path).
- Helps establish timeline of user activity.

6. Validate and Preserve

- Compare recovered files with hash values to confirm integrity.
- Store recovered evidence securely for analysis and presentation.

b. Write about malware handling in detail.

1. Definition

Malware handling refers to the systematic process of **identifying, isolating, analyzing, and documenting malicious software** during forensic investigations. The goal is to understand its behavior, preserve evidence, and prevent further damage.

2. Steps in Malware Handling

1. Identification

- Detect suspicious files, processes, or network activity.

- Use antivirus alerts, IDS/IPS logs, or anomaly detection tools.

2. Isolation

- Quarantine infected systems to prevent malware spread.
- Disconnect from the network and preserve volatile data (RAM dumps).

3. Collection & Preservation

- Acquire forensic images of infected drives.
- Collect malware samples, logs, registry entries, and system artifacts.
- Maintain chain of custody and hash values for integrity.

4. Static Analysis

- Examine malware without executing it.
- Study file headers, strings, and embedded code.
- Tools: IDA Pro, PE Explorer.

5. Dynamic Analysis

- Execute malware in a controlled sandbox environment.
- Observe behavior: file modifications, registry changes, network connections.
- Tools: Cuckoo Sandbox, Wireshark.

6. Reverse Engineering

- Disassemble or decompile malware to understand its logic.
- Identify payloads, encryption methods, and persistence mechanisms.

7. Documentation & Reporting

- Record findings: type of malware, infection vector, impact.
- Provide recommendations for removal and prevention.

Q3. [10 Marks each] - Answers

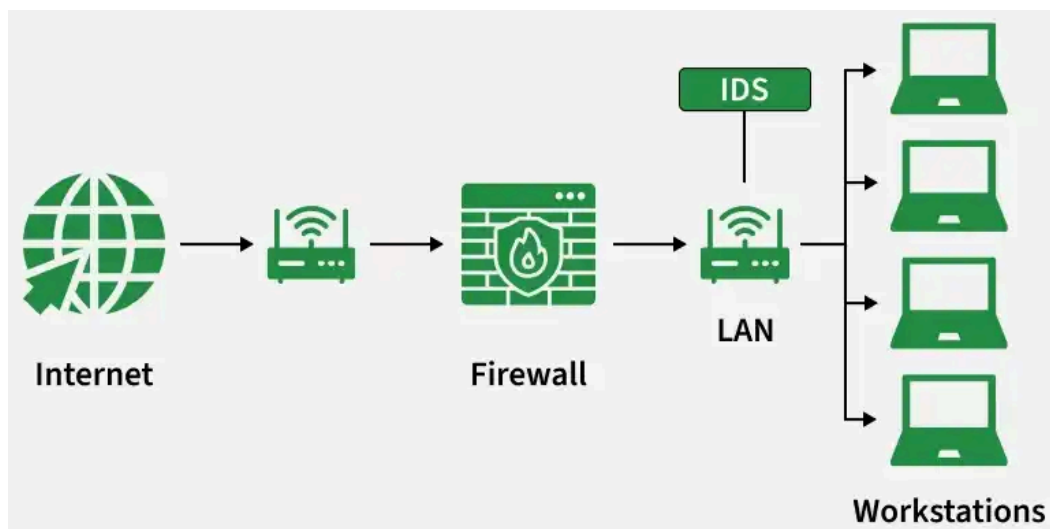
a. What is an Intrusion Detection System (IDS)? Explain.

What is an IDS?

- **Definition:** A software or hardware solution that inspects network packets and system logs for malicious activity.
- **Purpose:** To identify intrusions such as malware, unauthorized logins, or abnormal traffic patterns.
- **Function:** Detects suspicious activity and generates alerts for administrators to investigate.

How IDS Works

1. **Traffic Monitoring** – Captures and inspects data flowing across the network.
2. **Detection Methods:**
 - **Signature-based:** Matches traffic against known attack signatures (e.g., malware code patterns).
 - **Anomaly-based:** Uses machine learning or baselines of normal activity to flag deviations.
3. **Alerting** – Notifies administrators or feeds into a SIEM (Security Information and Event Management) system.
4. **Response** – Security teams investigate alerts and take manual action.



Types of IDS

Type	Function	Example Use Case
Network IDS (NIDS)	Monitors traffic across subnets and compares against attack signatures.	Detecting port scans or DDoS attempts.
Host IDS (HIDS)	Runs on individual devices, monitors system files and logs.	Detecting unauthorized file changes on a server.
Hybrid IDS	Combines NIDS and HIDS for broader coverage.	Enterprise-level monitoring across endpoints and networks

Real-World Example

- A **Network IDS** placed before a firewall can detect repeated login attempts from suspicious IPs, alerting administrators to a brute-force attack.
- A **Host IDS** on a critical server can flag unauthorized modifications to system files, indicating possible malware infection.

b. What is a file system and explain forensic analysis of a file system.

1. Definition of File System

A **file system** is the method and data structure an operating system uses to **store, organize, and manage files** on storage devices (hard drives, SSDs, USBs). It defines how data is named, accessed, and retrieved.

- Examples: **FAT32, NTFS (Windows), EXT3/EXT4 (Linux), HFS+ (MacOS)**.

2. Forensic Analysis of a File System

1. Identification

- Determine the type of file system (NTFS, FAT, EXT, etc.).
- Recognize potential sources of evidence (files, logs, metadata).

2. Preservation

- Create a **forensic image** (bit-by-bit copy) of the storage device.
- Generate hash values (MD5/SHA-1) to ensure integrity.

3. Metadata Examination

- Analyze file attributes (timestamps, permissions, ownership).
- Study Master File Table (MFT) in NTFS or inode structures in Linux.

4. File Recovery

- Recover deleted files from unallocated space.
- Use carving techniques to reconstruct files based on headers/footers.

5. Log & Journal Analysis

- Inspect system logs, journaling data, and registry entries.
- Helps reconstruct user/system activity timeline.

6. Hidden/Obscured Data Detection

- Identify hidden partitions, alternate data streams, or encrypted files.
- Detect steganography or obfuscation attempts.

7. Documentation & Reporting

- Record findings, tools used, and evidence recovered.
- Ensure admissibility in court.

Q4. [10 Marks each] - Answers

a. Explain in detail the task of a hardware forensic tool.

1. Definition

A **hardware forensic tool** is a specialized device or equipment used in digital forensics to **collect, preserve, and analyze evidence directly from physical hardware components** such as hard drives, memory chips, mobile devices, and network equipment. Unlike software tools, hardware tools ensure tamper-proof acquisition and maintain evidence integrity.

2. Tasks of a Hardware Forensic Tool

1. Write Blocking

- Prevents any modification to the original evidence during acquisition.
- Ensures investigators can only read data, not alter it.
- Example: Tableau Forensic Write Blocker.

2. Data Acquisition (Imaging)

- Creates a **bit-by-bit forensic image** of storage media.
- Captures deleted files, hidden partitions, and slack space.
- Example: Logicube Forensic Dossier.

3. Memory Chip Extraction

- Reads data directly from flash memory chips (NAND, EEPROM).
- Used in mobile forensics when devices are locked or damaged.
- Example: Chip-off technique with specialized readers.

4. Live Data Capture

- Collects volatile data (RAM contents, running processes) before shutdown.
- Useful in malware or intrusion investigations.
- Example: Hardware RAM capture devices.

5. Network Traffic Capture

- Dedicated hardware appliances intercept and record live traffic.
- Helps in **network forensics** and intrusion detection.
- Example: Network TAP devices.

6. Mobile Device Acquisition

- Extracts data from smartphones using specialized connectors.
- Supports SIM card cloning, JTAG extraction, and USB acquisition.
- Example: Cellebrite UFED hardware kit.

7. Evidence Preservation

- Ensures secure storage of forensic images and physical drives.
- Maintains chain of custody with tamper-proof seals and logs.

Q5. [10 Marks each] - Answers

a. Write a short note on different types of scanning tools.

1. Definition

Scanning tools are software utilities used in ethical hacking and forensics to **identify vulnerabilities, open ports, services, and potential security weaknesses** in systems and networks. They help investigators and administrators assess security posture.

2. Types of Scanning Tools

1. Port Scanners

- Detect open, closed, or filtered ports on a system.
- Example: **Nmap** – identifies services running on ports and OS details.

2. Vulnerability Scanners

- Scan systems for known vulnerabilities, misconfigurations, or missing patches.

- Example: **Nessus, OpenVAS** – report CVEs and risk levels.

3. Network Scanners

- Map devices connected to a network and analyze traffic.
- Example: **Angry IP Scanner, Advanced IP Scanner** – discover active hosts.

4. Web Application Scanners

- Test websites for issues like SQL injection, XSS, or insecure authentication.
- Example: **Acunetix, Burp Suite**.

5. Malware Scanners

- Detect malicious files or suspicious activity.
- Example: **ClamAV, Malwarebytes**.

6. Wireless Scanners

- Analyze Wi-Fi networks for weak encryption or rogue access points.
- Example: **Aircrack-ng, Kismet**.

b. Explain the different types of digital forensics.

1. Definition

Digital Forensics is the branch of forensic science that involves the identification, preservation, analysis, and presentation of digital evidence from electronic devices. It helps in investigating cybercrimes, fraud, and unauthorized access.

2. Types of Digital Forensics

1. Computer Forensics

- Focuses on desktops, laptops, and servers.
- Recovers deleted files, analyzes logs, and investigates malware.
- *Example:* Recovering deleted financial records from a suspect's PC.

2. Network Forensics

- Examines network traffic and communication logs.
- Detects intrusions, DDoS attacks, and unauthorized access.
- *Example:* Analyzing packet captures to trace a hacker's IP.

3. **Mobile Device Forensics**

- Extracts data from smartphones, tablets, and SIM cards.
- Recovers SMS, call logs, app data, and GPS locations.
- *Example:* Retrieving WhatsApp chats in a fraud investigation.

4. **Cloud Forensics**

- Investigates data stored in cloud platforms (Google Drive, iCloud).
- Deals with multi-tenant environments and remote evidence.
- *Example:* Accessing synced documents from a suspect's cloud account.

5. **Database Forensics**

- Analyzes structured data in databases.
- Detects unauthorized queries, data tampering, or insider threats.
- *Example:* Investigating manipulated entries in a banking database.

6. **Email Forensics**

- Examines email headers, metadata, and attachments.
- Traces sender identity and detects phishing or spoofing.
- *Example:* Tracking a phishing scam through header analysis.

7. **Memory Forensics**

- Extracts volatile data from RAM.
- Reveals running processes, encryption keys, and hidden malware.
- *Example:* Capturing RAM dumps to detect rootkits.

Q6. [10 Marks each] - Answers

a. Write a short note on: Illustrate investigation of a web browser.

1. Definition

Web browser investigation in digital forensics involves analyzing browser artifacts to uncover user activities such as visited websites, downloads, cached files, and stored credentials. Browsers often hold crucial evidence in cybercrime cases.

2. Key Areas of Investigation

1. Browsing History

- Records of visited websites, timestamps, and frequency.
- Helps reconstruct user activity timeline.

2. Cache & Temporary Files

- Stores images, scripts, and web content.
- May reveal accessed pages even if history is deleted.

3. Cookies

- Contain session data, login tokens, and user preferences.
- Useful for tracking authentication and online behavior.

4. Download Records

- Logs of files downloaded via the browser.
- Can link suspect to possession of illegal or incriminating files.

5. Form Data & Autofill

- Stores usernames, email addresses, and sometimes passwords.
- Provides insight into accounts accessed.

6. Bookmarks & Favorites

- Reveal frequently accessed or important sites.

7. Search Engine Queries

- Records of keywords searched by the user.
- Useful in profiling intent or criminal activity.

b. Write a short note on: Describe forensic duplication of a hard drive.

1. Definition

Forensic duplication of a hard drive is the process of creating an **exact bit-by-bit copy** of the entire storage media, including active files, deleted files, hidden partitions, and slack space. This duplicate is used for analysis while the original is preserved to maintain evidence integrity.

2. Steps in Forensic Duplication

1. Preparation

- Secure the original hard drive.
- Use **write blockers** to prevent accidental modification.

2. Imaging Process

- Create a forensic image (bit-by-bit copy) using tools like **FTK Imager, EnCase, dd, X-Ways**.
- Capture all sectors, including unallocated space.

3. Hashing for Integrity

- Generate hash values (MD5/SHA-1/SHA-256) before and after duplication.
- Ensures the duplicate is identical to the original.

4. Verification

- Compare hash values to confirm authenticity.
- Document the process for chain of custody.

5. Storage & Preservation

- Store the original drive securely.
- Use the duplicate for forensic analysis to avoid altering evidence.

c. Write a short note on: Explain the steps in router investigation.

1. Definition

Router investigation in digital forensics involves analyzing a router's configuration, logs, and traffic data to uncover evidence of unauthorized access, malicious activity, or policy violations. Routers often store valuable information about network usage and intrusions.

1. Secure and Preserve Router

Ensure the router is isolated to prevent tampering and preserve evidence.

- Disconnect router from external networks
- Document physical condition and connections
- Maintain chain of custody

2. Collect Configuration Data

Extract router settings and configurations for analysis.

- Export configuration files
- Note firewall rules, NAT settings, and access control lists
- Record firmware version

3. Analyze Router Logs

Examine logs to trace suspicious activity and intrusions.

- Review system logs for login attempts
- Check DHCP logs for connected devices
- Inspect traffic logs for anomalies

4. Examine Network Traffic

Identify malicious or unauthorized communication through packet analysis.

- Capture live traffic if possible
- Use tools like Wireshark for packet inspection
- Detect unusual patterns or IP addresses

5. Recover Deleted or Hidden Data

Investigate hidden or cleared logs that may contain evidence.

- Use forensic tools to recover deleted logs
- Check for hidden partitions or backup files

6. Document and Report Findings

Prepare a detailed forensic report for legal and organizational use.

- Summarize evidence collected
- Highlight suspicious activities
- Provide recommendations for security improvements

EHF-2024-December-PYQ Answers

Q1. [5 Marks each]

- (a) Explain cybercrime and its types.
- (b) Differentiate White, Black, and Grey Hat Hackers.
- (c) Discuss Digital Evidence and its types.
- (d) What is malware? Explain in brief the concept of a virus.

Q2. [10 Marks each]

- (a) What are the primary goals of ethical hacking and what are its phases?

Q3. [10 Marks each]

- (b) Explain the forensic duplication and investigation process in detail.

Q4. [10 Marks each]

- (b) Describe tasks in investigating e-mail crimes and violations.

Q5. [10 Marks each]

- (a) Explain the attacks on Network and Prevention.
- (b) What are computer forensics? Why is computer forensics important?

Q6. [5 Marks each]

Write a short note on (Any Four):

- (a) Firewall with Neat diagram.
- (e) Cyber Forensic tools.

Q1. [5 Marks each] - Answers

a. Explain cybercrime and its types.

Cybercrime refers to illegal activities carried out using computers, networks, or digital devices, and it is broadly classified into categories such as crimes against individuals, property, organizations, and society. These crimes range from identity theft and fraud to cyber terrorism and hacking.

Definition of Cybercrime

- **Cybercrime** is any criminal activity that involves computers, networks, or digital devices as tools, targets, or both.
- It includes acts like unauthorized access, data theft, online fraud, and disruption of services.

Types of Cybercrime

Category	Description	Examples
Against Individuals	Targets personal privacy, finances, or reputation.	Identity theft, phishing, cyberstalking, cyber defamation.
Against Organizations	Aimed at disrupting or stealing from businesses.	Corporate espionage, DDoS attacks, insider threats.
Against Society	Large-scale crimes affecting public order or national security.	Cyber terrorism, spreading fake news, financial fraud networks.

Common Examples of Cybercrime

- **Phishing scams:** Fake emails/websites tricking users into revealing sensitive data.
- **Ransomware attacks:** Encrypting files and demanding payment for decryption.
- **Data breaches:** Unauthorized access to confidential databases.
- **Cyber terrorism:** Using digital means to disrupt critical infrastructure.

b. Differentiate White, Black, and Grey Hat Hackers.

Aspect	White Hat Hackers	Black Hat Hackers	Grey Hat Hackers
Definition	Ethical hackers who test systems legally.	Malicious hackers who exploit systems illegally.	Hackers who operate between ethical and illegal.
Intent	Improve security and protect data.	Steal, damage, or misuse information.	May expose flaws without permission.
Legality	Legal, authorized by organizations.	Illegal, punishable by law.	Often unauthorized, legality is questionable.
Approach	Use same tools but report responsibly.	Exploit vulnerabilities secretly.	Identify flaws, sometimes disclose publicly.
Example	Penetration tester hired by a bank.	Hacker stealing credit card details.	Hacker revealing a bug on a website without consent.

c. Discuss Digital Evidence and its types.

Definition of Digital Evidence

- Digital evidence is any information stored or transmitted in digital form that can be used in investigations or legal proceedings.
- It is collected from computers, mobile devices, networks, or digital storage media.

Types of Digital Evidence

1. **Computer Files** – Documents, images, videos, or databases stored on a system.
2. **Email & Communication Records** – Emails, chat logs, or social media messages.
3. **Log Files** – System, server, or application logs showing user activity.
4. **Network Data** – Packets, IP addresses, or traffic analysis from networks.
5. **Mobile Evidence** – SMS, call records, app data from smartphones.

d. What is malware? Explain in brief the concept of a virus.

Definition of Malware

- Malware (malicious software) is any program or code designed to disrupt, damage, or gain unauthorized access to computer systems.
- It includes viruses, worms, trojans, ransomware, spyware, and other harmful software.

Concept of a Virus

- A **virus** is a type of malware that attaches itself to files or programs and spreads when those files are executed.
- It can corrupt data, slow down systems, or delete files.
- Viruses often require human action (like opening an infected file or email attachment) to spread.
- Example: A virus hidden in a document that, when opened, infects the system and replicates itself across other files.

Q2. [10 Marks each] - Answers

a. What are the primary goals of ethical hacking and what are its phases?

1. Primary Goals of Ethical Hacking

Ethical hacking is performed by authorized professionals to strengthen security. Its main goals are:

1. **Identify Vulnerabilities** – Discover weaknesses before malicious hackers exploit them.
2. **Strengthen Security** – Provide recommendations to patch flaws.
3. **Prevent Attacks** – Simulate real-world threats to test defenses.
4. **Ensure Compliance** – Help organizations meet legal and industry standards.
5. **Protect Data & Assets** – Safeguard sensitive information from theft or misuse.

6. **Improve Incident Response** – Train organizations to respond effectively to breaches.

2. Phases of Lifecycle

1. Reconnaissance (Information Gathering)

- Collect information about the target (IP addresses, domains, DNS records).
- Types: Passive (public sources) and Active (direct probing).

2. Scanning

- Identify live hosts, open ports, and services.
- Tools: Nmap, Nessus.
- Purpose: Detect potential entry points.

3. Gaining Access

- Exploit vulnerabilities to enter the system.
- Tools: Metasploit, SQL injection scripts.
- Goal: Demonstrate how attackers could compromise systems.

4. Maintaining Access

- Install backdoors or create persistence to show long-term risks.
- Used to test how attackers can remain undetected.

5. Reporting

- Document findings, exploited vulnerabilities, tools used, and recommendations.
- Provides actionable steps for organizations to fix weaknesses.

Q3. [10 Marks each] - Answers

b. Explain the forensic duplication and investigation process in detail.

1. Forensic Duplication

Forensic duplication is the **creation of an exact bit-by-bit copy** of digital storage media (hard drives, SSDs, USBs, memory cards). It ensures that investigators analyze the duplicate while the original remains untouched, preserving evidence integrity.

Steps in Forensic Duplication:

1. Preparation

- Secure the original device.
- Use **write blockers** to prevent accidental modification.

2. Imaging

- Create a forensic image using tools like **FTK Imager, EnCase, dd, X-Ways**.
- Capture all sectors, including deleted files and slack space.

3. Hashing

- Generate hash values (MD5/SHA-1/SHA-256) before and after duplication.
- Confirms the duplicate is identical to the original.

4. Verification

- Compare hash values to validate authenticity.
- Document the process for chain of custody.

5. Storage

- Preserve the original securely.
- Use the duplicate for analysis and reporting.

2. Forensic Investigation Process

Once duplication is complete, investigators analyze the forensic image to uncover evidence.

Steps in Investigation:

1. Identification

- Recognize potential evidence sources (files, logs, metadata).

2. Preservation

- Ensure evidence integrity with proper documentation and chain of custody.

3. Collection

- Extract relevant data (emails, documents, logs, deleted files).

4. Examination

- Study file systems, metadata, registry entries, and hidden partitions.
- Recover deleted or hidden data.

5. Analysis

- Correlate findings with timelines.
- Identify user activity, malware traces, or unauthorized access.

6. Documentation

- Record tools used, steps taken, and evidence recovered.

7. Presentation

- Prepare a structured forensic report.
- Present findings in court with expert testimony.

Q4. [10 Marks each] - Answers

b. Describe tasks in investigating e-mail crimes and violations.

1. Definition

E-mail forensics is the process of examining e-mails to detect fraud, cybercrime, harassment, phishing, or policy violations. It focuses on analyzing content, metadata, and headers to trace the sender's identity and verify authenticity.

2. Key Tasks in Investigation

1. Header Analysis

- Examine "Received" fields to trace the sender's IP.
- Check "From," "Reply-To," "Message-ID," and "Return-Path" for spoofing.

2. Content Examination

- Analyze body text, attachments, and embedded links.
- Detect phishing attempts, malware payloads, or fraudulent messages.

3. Tracing Sender Identity

- Perform IP lookup (WHOIS, geolocation).
- Correlate with server logs and timestamps.

4. Attachment & Malware Analysis

- Inspect suspicious attachments for malicious code.
- Use sandboxing tools to observe behavior.

5. Authentication Checks

- Verify SPF, DKIM, and DMARC records to confirm legitimacy.

6. Correlation with Logs

- Match e-mail activity with system logs, firewall records, or user login data.

7. Documentation & Reporting

- Record findings, tools used, and evidence recovered.
- Maintain chain of custody for legal admissibility.

Q5. [10 Marks each] - Answers

a. Explain the attacks on Network and Prevention.

1. Definition

A **network attack** is any unauthorized activity aimed at disrupting, stealing, or damaging data and resources within a computer network. Attackers exploit vulnerabilities to gain access, cause downtime, or steal sensitive information.

2. Common Types of Network Attacks

1. Denial of Service (DoS) / Distributed DoS (DDoS)

- Flooding the network with traffic to make services unavailable.

2. Man-in-the-Middle (MITM)

- Attacker intercepts communication between two parties to steal or alter data.

3. Phishing & Social Engineering

- Trick users into revealing credentials or sensitive data.

4. Sniffing / Eavesdropping

- Capturing unencrypted data packets traveling across the network.

5. Password Attacks (Brute Force, Dictionary, Keylogging)

- Attempting to crack user credentials to gain unauthorized access.

6. Malware-based Attacks (Viruses, Worms, Trojans)

- Malicious software spreads through the network, damaging systems.

3. Prevention Techniques

1. Firewalls & Intrusion Detection/Prevention Systems (IDS/IPS)

- Monitor and filter malicious traffic.

2. Encryption (SSL/TLS, VPNs)

- Protects data in transit from sniffing and MITM attacks.

3. Strong Authentication

- Use multi-factor authentication and strong password policies.

4. Regular Patching & Updates

- Fix vulnerabilities in operating systems and applications.

5. User Awareness & Training

- Educate users about phishing and social engineering.

6. Monitoring & Logging

- Continuous monitoring of traffic and system logs for anomalies.

7. Backup & Recovery Plans

- Ensure data can be restored after ransomware or DoS attacks.

b. What are computer forensics? Why is computer forensics important?

1. Definition

Computer Forensics is a branch of digital forensics that deals with the **identification, preservation, analysis, and presentation of evidence from computer systems** in a legally admissible manner. It involves recovering deleted files, analyzing logs, detecting malware, and reconstructing user activity to support investigations.

2. Importance of Computer Forensics

1. Crime Investigation

- Helps law enforcement investigate cybercrimes such as hacking, fraud, identity theft, and cyber terrorism.

2. Evidence Preservation

- Ensures digital evidence is collected and stored without alteration, maintaining integrity for court proceedings.

3. Legal Admissibility

- Provides structured reports and expert testimony that can be accepted in courts of law.

4. Corporate Security

- Assists organizations in detecting insider threats, data breaches, and policy violations.

5. Incident Response

- Enables quick detection and analysis of security incidents, minimizing damage.

6. Data Recovery

- Recovers deleted, hidden, or encrypted files that may serve as crucial evidence.

7. Deterrence of Cybercrime

- Acts as a preventive measure by strengthening systems against future attacks.

Q6. [10 Marks each] - Answers

a. Write a short note on Firewall

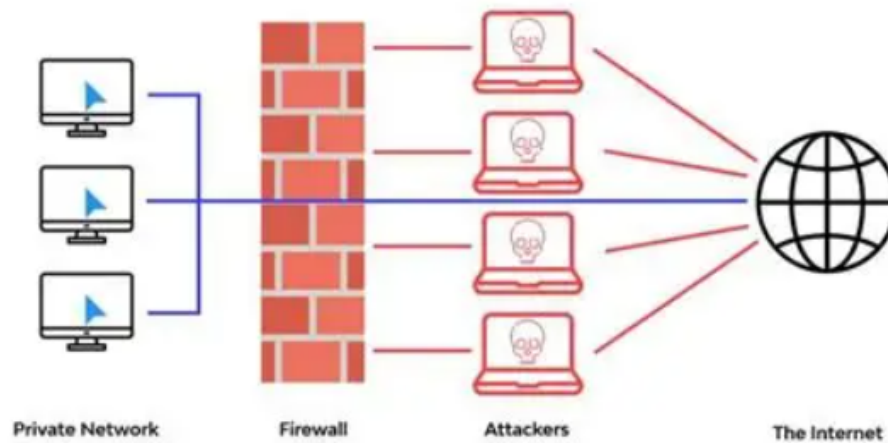
1. Definition

A **firewall** is a security system (hardware or software) that monitors and controls incoming and outgoing network traffic based on predefined rules. It acts as a barrier between a trusted internal network and untrusted external networks (like the Internet).

2. Functions of a Firewall

1. **Traffic Filtering** – Allows or blocks data packets based on rules.
2. **Access Control** – Restricts unauthorized users from accessing private networks.
3. **Monitoring & Logging** – Records traffic for analysis and forensic investigation.
4. **Protection Against Attacks** – Helps prevent malware, DoS, and unauthorized access.
5. **Policy Enforcement** – Ensures compliance with organizational security policies.

What is a Firewall



3. Types of Firewalls

- **Packet-Filtering Firewall** → Examines headers of packets.
- **Stateful Inspection Firewall** → Tracks active connections and traffic state.
- **Proxy Firewall** → Acts as an intermediary between users and services.
- **Next-Generation Firewall (NGFW)** → Includes intrusion prevention, deep packet inspection, and application control.

EHF-2023-May-PYQ Answers

Q1. [5 Marks each]

- (a) Explain Mobile forensic. What are various challenges in mobile forensics?
- (b) Explain Forensic Duplicates as Admissible Evidence.
- (c) What is the evidence handling procedure?
- (d) What are Challenges in network forensics?

Q2. [10 Marks each]

- (b) Compare active attacks vs. passive attacks. Classify cybercrimes and explain any one briefly.

Q3. [10 Marks each]

- (a) Discuss basic security precautions to be taken to safeguard Laptops and wireless devices. What are the devices related to security issues?
- (b) Explain Volatile Data Collection from a Windows system.

Q4. [10 Marks each]

- (a) What do you understand by social engineering? Give its classification.
- (b) Briefly explain Types of digital Evidence with examples.

Q5. [10 Marks each]

- (a) Explain the process for collecting Network Based Evidence.

Q6. [10 Marks each]

Write a short note on (Any Two):

- 3. Email Tracing - Internet Fraud.

Q1. [5 Marks each] - Answers

a. Explain Mobile forensic. What are various challenges in mobile forensics?

Definition of Mobile Forensics

- Mobile forensics is a branch of digital forensics that deals with the recovery, preservation, and analysis of digital evidence from mobile devices (smartphones, tablets, SIM cards, memory cards).
- It is used in criminal investigations, cybercrime cases, and legal proceedings.

Challenges in Mobile Forensics

1. **Device Diversity** – Wide variety of operating systems (Android, iOS, Windows) and hardware models.
2. **Rapid Updates** – Frequent OS and app updates make forensic tools outdated quickly.
3. **Data Encryption** – Strong encryption and security features hinder evidence extraction.
4. **Volatile Data** – Mobile data (RAM, temporary files, app caches) can be lost easily.
5. **Legal & Privacy Issues** – Accessing personal data requires strict adherence to laws and warrants.

b. Explain Forensic Duplicates as Admissible Evidence.

Forensic Duplicates as Admissible Evidence

- A **forensic duplicate** is an exact bit-by-bit copy of digital storage media created during forensic investigation.
- It preserves all data, including hidden, deleted, or system files, ensuring the original evidence remains untouched.
- Courts accept forensic duplicates as admissible evidence because they maintain the **authenticity, integrity, and reliability** of digital data.

Importance in Admissibility

1. **Authenticity** – Duplicate is identical to the original, ensuring genuineness.
2. **Integrity** – Original evidence remains unaltered, protecting chain of custody.
3. **Reproducibility** – Investigators can analyze duplicates multiple times without risk.
4. **Legal Acceptance** – Courts recognize forensic images if proper procedures are followed.
5. **Comprehensive Evidence** – Includes deleted or hidden files that may be crucial in cases.

c. What is the evidence handling procedure?

Evidence Handling Procedure

Evidence handling in digital forensics refers to the systematic process of collecting, preserving, analyzing, and presenting digital evidence while ensuring its integrity and admissibility in court.

Steps in Evidence Handling

1. **Identification** – Recognize potential sources of digital evidence (computers, mobiles, storage devices).
2. **Collection** – Securely acquire evidence using forensic tools without altering original data.
3. **Preservation** – Store evidence in a safe environment, maintaining integrity and chain of custody.
4. **Documentation** – Record every step (who collected, when, how) for accountability.
5. **Analysis** – Examine forensic copies to extract relevant information.
6. **Presentation** – Present findings in court with proper reports and expert testimony.

d. What are Challenges in network forensics?

Challenges in Network Forensics

Network forensics involves monitoring, capturing, and analyzing network traffic to investigate cyber incidents. However, several challenges make it complex:

1. **High Volume of Data** – Networks generate massive traffic, making storage and analysis difficult.
2. **Real-Time Monitoring** – Attacks happen quickly; forensic tools must capture evidence instantly.
3. **Encryption** – Widespread use of SSL/TLS and VPNs hides malicious activity.
4. **Dynamic IPs & NAT** – Changing IP addresses and shared networks complicate attacker identification.
5. **Distributed Attacks** – DDoS and botnet attacks originate from multiple sources, making tracing harder.
6. **Legal & Privacy Issues** – Monitoring network traffic must comply with laws and user privacy rights.

Q2. [10 Marks each] - Answers

a. Compare active attacks vs. passive attacks. Classify cybercrimes and explain any one briefly.

Aspect	Active Attacks	Passive Attacks
Definition	Attacker attempts to alter, disrupt, or damage system resources.	Attacker silently monitors or intercepts data without altering it.
Nature	Aggressive, intrusive, and directly harmful.	Stealthy, non-intrusive, and hidden.
Goal	Gain unauthorized access, modify data, or disrupt services.	Collect information for later exploitation.
Examples	DoS/DDoS, SQL Injection, Malware injection, Password cracking.	Packet sniffing, traffic analysis, eavesdropping.

Aspect	Active Attacks	Passive Attacks
Detection	Easier to detect due to visible disruption.	Harder to detect since no changes are made to data flow.
Impact	Immediate damage (service downtime, data loss, system compromise).	Long-term risk (information leakage, espionage).
Prevention	Firewalls, IDS/IPS, encryption, patching, strong authentication.	Encryption, secure protocols (SSL/TLS, VPN), monitoring suspicious logs.

Classification of Cybercrimes

Cybercrimes can be classified into:

1. **Crimes against Individuals** → Identity theft, cyberstalking, online harassment.
2. **Crimes against Property** → Hacking, data theft, malware attacks.
3. **Crimes against Organizations** → Insider threats, corporate espionage, ransomware.
4. **Crimes against Society** → Cyber terrorism, spreading fake news, child exploitation.

Example Explained – Identity Theft (Crime against Individuals)

- **Definition:** Unauthorized use of someone's personal information (like Aadhaar number, credit card details, or login credentials) to commit fraud.
- **Process:** Attackers steal data via phishing emails, malware, or breaches.
- **Impact:** Victims face financial loss, damaged credit scores, and privacy violations.
- **Case Example:** The **Equifax data breach (2017)** exposed millions of users' personal data, leading to widespread identity theft.

Q3. [10 Marks each] - Answers

a. Discuss basic security precautions to be taken to safeguard Laptops and wireless devices. What are the devices related to security issues?

1. Security Precautions

1. Strong Authentication

- Use complex passwords and enable multi-factor authentication (MFA).

2. Regular Updates & Patching

- Keep operating systems, browsers, and applications updated to fix vulnerabilities.

3. Firewalls & Antivirus

- Enable built-in firewalls and install reputable antivirus/anti-malware software.

4. Data Encryption

- Encrypt sensitive files and enable full-disk encryption (e.g., BitLocker, FileVault).

5. Secure Wi-Fi Usage

- Connect only to trusted networks; avoid public Wi-Fi or use VPNs.

6. Physical Security

- Lock laptops with cable locks; avoid leaving devices unattended.

7. Backup & Recovery

- Maintain regular backups to external drives or secure cloud storage.

2. Devices Related to Security Issues

Certain devices are more prone to security risks due to connectivity and portability:

- **Laptops** → Susceptible to theft, malware infections, and unauthorized access.
- **Smartphones & Tablets** → Store sensitive data, vulnerable to app-based malware and SIM cloning.

- **Wireless Routers & Access Points** → Weak encryption or misconfiguration can expose networks.
- **USB Drives & External Storage** → Can spread malware or be used for data theft.
- **IoT Devices (Smart TVs, Cameras, Wearables)** → Often lack strong security, making them easy targets.

b. Explain Volatile Data Collection from a Windows system.

1. Definition

Volatile data refers to information stored in temporary memory (RAM) that is lost once the system is powered off. Collecting volatile data from a Windows system is crucial in forensic investigations because it may contain evidence of **running processes, network connections, encryption keys, and logged-in users**.

2. Steps in Volatile Data Collection

1. Preparation

- Use trusted forensic tools.
- Ensure chain of custody documentation.

2. Identify Running Processes

- Commands: `tasklist` , `wmic process list` .
- Tools: **PsList, Volatility Framework**.

3. Capture Network Connections

- Commands: `netstat -ano` , `ipconfig /all` .
- Reveals active sessions, IP addresses, and ports.

4. Logged-in Users & Sessions

- Commands: `query user` , `whoami` .
- Identifies current user activity.

5. System Information

- Commands: `systeminfo` , `wmic os get caption` .
- Provides OS version, uptime, and patch details.

6. Clipboard & Temporary Data

- Tools: **NirSoft utilities** can capture clipboard contents.

7. Memory Dump

- Tools: **WinDbg, DumpIt, Belkasoft RAM Capturer**.
- Creates a snapshot of RAM for later analysis.

8. Hashing & Preservation

- Generate hash values of collected data.
- Store securely for integrity verification.

Q4. [10 Marks each] - Answers

a. What do you understand by social engineering? Give its classification.

1. Definition

Social engineering is a psychological manipulation technique used by attackers to trick individuals into revealing confidential information or performing actions that compromise security. Instead of exploiting technical flaws, it exploits **human trust, curiosity, and fear**.

2. Classification of Social Engineering

1. Phishing

- Fraudulent emails or websites designed to steal credentials.
- Example: Fake bank email asking for login details.

2. Spear Phishing

- Targeted phishing aimed at specific individuals or organizations.
- Example: Personalized email to a company executive.

3. Vishing (Voice Phishing)

- Attackers use phone calls to trick victims into sharing sensitive data.
- Example: Caller pretending to be from a bank's fraud department.

4. Smishing (SMS Phishing)

- Fraudulent text messages with malicious links.
- Example: Fake delivery SMS asking to click a link.

5. Baiting

- Offering something enticing (like free software or USB drives) to lure victims.
- Example: Infected USB labeled "Salary Data" left in office.

6. Tailgating / Piggybacking

- Attacker physically follows authorized personnel into restricted areas.
- Example: Pretending to be a delivery person to enter secure premises.

b. Briefly explain Types of digital Evidence with examples.

1. Definition

Digital evidence refers to any information stored or transmitted in digital form that can be used in investigations or legal proceedings. It is collected from computers, mobile devices, networks, and other electronic sources.

2. Types of Digital Evidence

1. Computer Files

- Documents, spreadsheets, images, videos, or databases stored on a system.
- *Example:* A deleted Excel sheet showing fraudulent transactions.

2. Email & Communication Records

- Emails, chat logs, instant messages, or social media conversations.

- *Example:* Phishing emails traced to a scammer's account.

3. Log Files

- System, server, or application logs that record user activity.
- *Example:* Login attempts recorded in Windows Event Viewer.

4. Network Data

- Packet captures, IP addresses, and traffic analysis.
- *Example:* Tracing a hacker's IP through captured network packets.

5. Mobile Evidence

- SMS, call records, app data, GPS locations from smartphones.
- *Example:* WhatsApp chats retrieved in a fraud investigation.

6. Cloud Evidence

- Data stored on cloud platforms (Google Drive, iCloud, OneDrive).
- *Example:* Documents synced to a suspect's cloud account.

7. Database Evidence

- Structured records from databases.
- *Example:* Manipulated entries in a banking database.

Q5. [10 Marks] - Answers

a. Explain the process for collecting Network Based Evidence.

1. Definition

Network-based evidence refers to digital traces collected from network traffic, logs, and communication channels that help investigators identify unauthorized access, cyberattacks, or suspicious activity. It is crucial in **network forensics** for detecting intrusions and reconstructing events.

2. Steps in Collecting Network-Based Evidence

1. Preparation & Legal Authorization

- Ensure proper warrants/permissions.
- Maintain chain of custody from the start.

2. **Identify Sources of Evidence**

- Routers, firewalls, intrusion detection systems (IDS/IPS).
- Server logs, packet captures, proxy logs.

3. **Traffic Capture**

- Use tools like **Wireshark, tcpdump, NetFlow analyzers**.
- Collect live packet data including headers, payloads, and timestamps.

4. **Log Collection**

- Gather logs from routers, firewalls, mail servers, and applications.
- Helps trace IP addresses, login attempts, and suspicious activity.

5. **Preservation of Evidence**

- Store captured packets and logs securely.
- Generate hash values to ensure integrity.

6. **Analysis**

- Examine traffic patterns, anomalies, and suspicious connections.
- Detect malware communication, DoS attempts, or unauthorized access.

7. **Correlation**

- Link network evidence with system logs, user activity, and timestamps.
- Builds a timeline of events.

8. **Documentation & Reporting**

- Record tools used, data collected, and findings.
- Prepare reports for legal admissibility.

Q6. [10 Marks each] - Answers

a. Write a short note on: Email Tracing - Internet Fraud.

1. Definition

Email tracing is a forensic technique used to track the origin of an email by analyzing its **headers, routing information, and metadata**. It helps investigators identify the sender's IP address, mail server, and possible location. This is crucial in detecting **internet fraud**, where attackers use fake or spoofed emails to deceive victims.

2. Steps in Email Tracing

1. **Access Full Header** – Extract complete header details from the email.
2. **Analyze "Received" Fields** – Identify the original sending IP from the earliest entry.
3. **Check "From" & "Reply-To" Fields** – Detect spoofing or mismatched addresses.
4. **Inspect "Message-ID" & "Return-Path"** – Verify authenticity of the sending server.
5. **Perform IP Lookup** – Use WHOIS or geolocation tools to trace sender's ISP and location.
6. **Correlate with Logs** – Match email activity with server logs and timestamps.

3. Role in Internet Fraud Investigation

- Fraudsters often use **phishing emails** to steal credentials or financial data.
- Email tracing helps investigators **unmask fake identities**, locate compromised servers, and gather evidence admissible in court.
- Example: In a **phishing scam**, tracing the header reveals the attacker's IP, linking the fraud to a server in another country.